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MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

Linux OS and Scripting

CYBERSECURITY LOG ANALYSIS USING LINUX TOOLS

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CYBERSECURITY LOG ANALYSIS USING LINUX TOOLS

1. Selected Cybersecurity Scenario

Scenario: Detection of SSH Brute-Force Login Attempts

An SSH brute-force attack is a type of cyberattack in which an attacker repeatedly attempts to log in to a system by guessing passwords. Attackers may try different usernames and passwords to gain unauthorized access.

Linux authentication logs record successful and failed login attempts. Analysing these logs helps identify repeated login failures and suspicious activities.

Security Significance:

- Helps detect unauthorized login attempts.
- Identifies IP addresses generating repeated login failures.
- Helps identify accounts targeted by attackers.
- Supports the implementation of security measures to prevent unauthorized access.

2. Log File Acquisition

A simulated authentication log file was created for this assignment. The simulation represents SSH login activity on a Linux system and contains both successful and failed login attempts.

Environment and Tools Used:

- Operating System: Kali Linux
- Log File: auth.log
- Tools: Bash, grep, sed, awk, sort, and uniq

Procedure:

1. Created a working directory in Kali Linux.
2. Created a sample authentication log file named auth.log.
3. Added simulated SSH login records containing successful and failed authentication attempts.

```
┌──(praneetha㉿Praneetha)-[~/log-analysis]
└─$ cat auth.log
Nov 02 10:15:21 kali sshd[1234]: Failed password for invalid user admin from 192.168.1.50 port 54321 ssh2
Nov 02 10:15:25 kali sshd[1235]: Failed password for invalid user root from 192.168.1.50 port 54322 ssh2
Nov 02 10:15:29 kali sshd[1236]: Failed password for invalid user test from 192.168.1.50 port 54323 ssh2
Nov 02 10:15:33 kali sshd[1237]: Failed password for invalid user guest from 192.168.1.50 port 54324 ssh2
Nov 02 10:15:37 kali sshd[1238]: Failed password for invalid user admin from 192.168.1.50 port 54325 ssh2
Nov 02 10:16:01 kali sshd[1240]: Failed password for student from 192.168.1.25 port 54326 ssh2
Nov 02 10:16:05 kali sshd[1241]: Failed password for student from 192.168.1.25 port 54327 ssh2
Nov 02 10:17:10 kali sshd[1242]: Accepted password for student from 192.168.1.10 port 54328 ssh2
Nov 02 10:18:20 kali sshd[1243]: Failed password for invalid user guest from 192.168.1.50 port 54329 ssh2
```

4. Analysed the records using Linux command-line tools and a Bash script.

The log file contains simulated data and does not include real passwords or confidential information.

3. Structure of the Log File

The authentication log contains information about SSH login attempts.

The important fields are:

1. **Timestamp:** Indicates the date and time of the event.
2. **Hostname:** Identifies the system that generated the log.
3. **Process Name:** Identifies the service generating the event, such as sshd.
4. **Process ID (PID):** Identifies the running process.
5. **Event Message:** Describes the authentication event.
6. **Username:** Indicates the account used in the login attempt.
7. **IP Address:** Identifies the source address of the login attempt.

Example Log Entry:

Nov 02 10:15:21 kali sshd[1234]: Failed password for invalid user admin from 192.168.1.50 port 54321 ssh2

This entry indicates a failed SSH login attempt for the invalid username admin from the IP address 192.168.1.50.

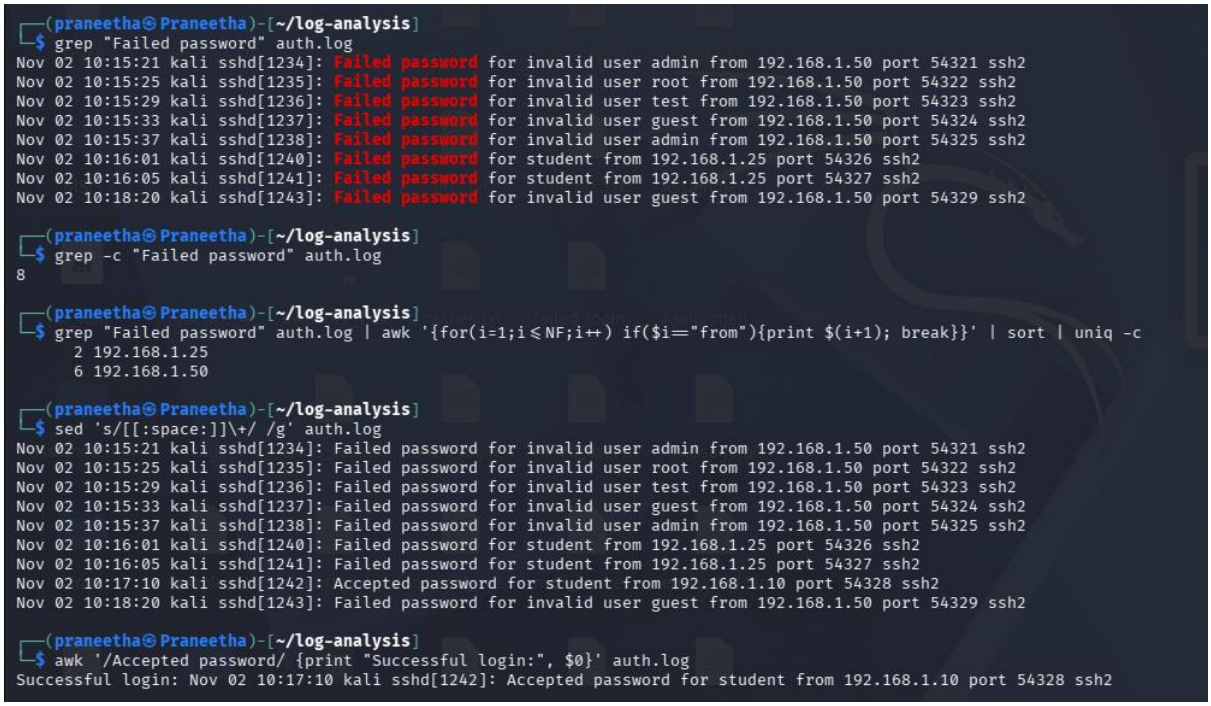
4. Cybersecurity Investigation Questions

The following questions were formulated for the investigation:

1. Which IP addresses generated failed SSH login attempts?

2. Which IP addresses generated three or more failed login attempts?
3. How many successful login attempts were recorded?
4. Were multiple usernames targeted from the same IP address?
5. Which IP addresses require further security investigation?

5. Linux Commands Used for Log Analysis



5.1 Power Filters and Pipelines

Filters and pipelines are used to extract relevant information from log files and process the output of commands.

Command 1: Display failed login attempts

grep "Failed password" auth.log

Purpose: Displays only the records containing failed password attempts.

Command 2: Count failed login attempts

grep -c "Failed password" auth.log

Purpose: Counts the total number of failed password records.

Command 3: Extract and count IP addresses

```
grep "Failed password" auth.log |
awk '{for(i=1;i<=NF;i++) if($i=="from"){print $(i+1); break}}' |
sort | uniq -c
```

Purpose:

- grep selects failed login records.
- awk extracts the IP address following the word from.
- sort arranges identical IP addresses together.
- uniq -c counts the occurrences of each IP address.

5.2 Using the sed Command

```
sed 's/[[:space:]]\+/ /g' auth.log
```

Purpose: Replaces consecutive whitespace characters with a single space, making the log records easier to process.

5.3 Using the awk Command

```
awk '/Accepted password/ {print "Successful login:", $0}' auth.log
```

Purpose: Identifies and displays log records containing successful SSH password authentication.

6. Bash Script for Log Analysis

The following Bash script integrates filters, pipelines, sed, and awk to analyse the authentication log.

File Name: analyze.sh

```
#!/bin/bash

LOG_FILE="auth.log"
THRESHOLD=3

if [ ! -f "$LOG_FILE" ]; then
    echo "Error: Log file not found."
    exit 1
fi

echo "SSH LOG ANALYSIS"

sed 's/[[:space:]]\+/ /g' "$LOG_FILE" > cleaned.log

FAILED=$(grep -c "Failed password" cleaned.log)
SUCCESS=$(grep -c "Accepted password" cleaned.log)

echo "Total failed login attempts: $FAILED"
echo "Total successful login attempts: $SUCCESS"

echo
echo "Failed login attempts by IP address:"

grep "Failed password" cleaned.log |
awk '{for(i=1;i<=NF;i++) if($i=="from"){print $(i+1); break}}' |
sort | uniq -c

echo
echo "Suspicious IP addresses (3 or more failures):"

grep "Failed password" cleaned.log |
awk '{for(i=1;i<=NF;i++) if($i=="from"){print $(i+1); break}}' |
sort | uniq -c |
awk -v limit="$THRESHOLD" \
'$1 >= limit {print $2, "-", $1, "failed attempts"}'

echo
echo "ANALYSIS COMPLETE"
```

7. Explanation of the Bash Script

1. **Shebang (#!/bin/bash):** Specifies Bash as the interpreter.
2. **Variable declaration:** Stores the log filename and suspicious activity threshold.
3. **File validation:** Checks whether the input log file exists before processing.

4. **sed command:** Normalizes whitespace and saves the processed records in cleaned.log.
5. **grep command:** Filters failed and successful login records and counts their occurrences.
6. **awk command:** Extracts the source IP address from each failed login record.
7. **sort command:** Groups identical IP addresses together.
8. **uniq command:** Counts the number of occurrences of each IP address.
9. **Threshold comparison:** Identifies IP addresses with three or more failed attempts.
10. **echo command:** Displays the analysis results in a readable format.

8. Execution Procedure

The script can be executed in Kali Linux using the following steps.

Step 1: Give execution permission to the script.

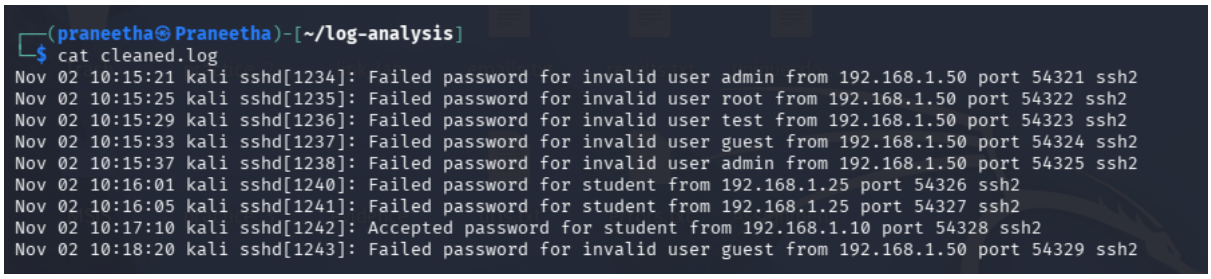
chmod +x analyze.sh

Step 2: Execute the script.

./analyze.sh

Step 3: View the generated output file.

cat cleaned.log



The input file auth.log and the script analyze.sh must be present in the same working directory.

9. Output and Identification of Suspicious Activities

Using the sample log file, the expected output is:

```
┌──(praneetha㉿Praneetha)-[~/log-analysis]
└─$ ./analyze.sh
SSH LOG ANALYSIS
Total failed login attempts: 8
Total successful login attempts: 1

Failed login attempts by IP address:
      2 192.168.1.25
      6 192.168.1.50

Suspicious IP addresses (3 or more failures):
192.168.1.50 - 6 failed attempts

ANALYSIS COMPLETE
```

Observations:

1. A total of eight failed login attempts were recorded.
2. One successful login attempt was recorded.
3. The IP address 192.168.1.50 generated six failed login attempts.
4. The IP address 192.168.1.25 generated two failed login attempts.
5. The address 192.168.1.50 exceeded the selected threshold and was classified as suspicious.

The repeated attempts involving different usernames from 192.168.1.50 may indicate possible brute-force activity. However, further investigation is required to confirm whether the activity is malicious.

10. Justification of the Threshold

A threshold of three or more failed login attempts per IP address was selected to identify potentially suspicious activity.

The reasons for selecting this threshold are:

1. Repeated login failures may indicate password guessing.
2. The threshold helps distinguish repeated attempts from a small number of isolated failures.
3. It is simple to implement and suitable for the simulated dataset.

4. The threshold is illustrative and should be adjusted according to the security requirements and normal login behaviour of an actual system.

An IP address exceeding the threshold is flagged for investigation rather than automatically classified as malicious.

11. False Positives and Limitations

11.1 False Positives

False positives occur when legitimate activities are incorrectly identified as suspicious.

- A genuine user may enter an incorrect password several times.
- Multiple users may share the same public IP address.
- Automated services may repeatedly attempt authentication using outdated credentials.
- An authorized administrator may generate repeated failures while testing a system.

11.2 Limitations

- The analysis uses simulated data rather than real authentication logs.
- The script counts failures per IP address but does not consider the time interval between attempts.
- It does not correlate events from multiple systems.
- It does not verify whether a successful login was legitimate.
- Attackers using multiple IP addresses may avoid detection by remaining below the threshold.

Therefore, the results provide an initial indication of suspicious activity and should be supported by further investigation.

12. Conclusion

This assignment demonstrates the use of Linux command-line tools for cybersecurity log analysis. The combination of Bash scripting, filters, pipelines, sed, and awk enables automated processing of authentication logs.

The analysis identified an IP address with six failed login attempts, exceeding the selected threshold of three attempts. This activity may indicate a possible SSH brute-force attack and requires further investigation.

The experiment demonstrates how log analysis can help security administrators identify suspicious authentication patterns and improve system security.

13. References and Assistance

1. GNU Grep Manual: https://www.gnu.org/software/grep/manual/grep.html
2. GNU Sed Manual: https://www.gnu.org/software/sed/manual/sed.html
3. GNU Awk User's Guide: https://www.gnu.org/software/gawk/manual/gawk.html
4. GNU Bash Reference Manual: https://www.gnu.org/software/bash/manual/